Adaptive Detection of Advanced Persistent Threats (APT) with Graph Neural Networks and Rehearsal-Based Continual Learning on Wazuh EDR Telemetry.

Purpose of the Paper:-

The main purpose of this Paper is to improve the detection of Advanced Presistent Threats in cyber - Security Systems.

APTs are Stealthy, long-term attacks that infiltrate Systems, escalate privilages, and remain hidden for long perods.

* Traditional detection methods struggle because:

→ Signature-based Systems detect only known attacks.

→ Static machine learning models fail when attack Patterns change

→ Concept drift occur when attack behaviour evolve.

→ Catastrophic forgetting happens when models forget Previously learned threats during updates.

* To address these issues, the paper propose an adpative APT detection framework that:

→ uses graphs neural Network (GNN) to model relati -onships between System entities.

→ Applies Rehearsal-based Continual Learning so the model can learn new threats without forgetting old ones.

→ uses Wazuh EDR telemetry data from endpoints to detect malicious behaviour.

* The goal of the Prj is to create a Scalable, adaptive and resilient CyberSecurity detection System Suitable for Security Operation Centers (SOC) and Endpoint Detection & Respon -se (EDR) environments.

Key Contributions :-

1. Graph based threat representation.

→ It converts endpoint telemetry data into graphs.

→ Captures behavioural relationships between components which traditional feature-based methods miss.

2. Integration of Graph neural network :-

→ The study applies GNN to learn patterns from dense interaction graphs, enabling the model to detect complex multi-stage attacks like APTs.

3. Continual Learning for Evolving Attacks :-

→ The paper integrates rehersal-based continual learning, allowing the model to :

- Learn new attack behaviors
- Retain knowledge of previous attack
- Handle concept drift.

4. Real World Telemetry Evaluation.

→ Fram work is evaluated using :

- Wbith EDR telemetry
- Sandbox-executed APT attack scenarios.

* The system achived f1-Score of 0.98, outperforming tradit-onal static models.

5. Practical SOC/EDC Deployment potential :-

The proposed framework is designed to be deployable in real Cyber Security env's, especially in :

- SOC monitoring Systems
- Endpoint detection platforms
- SIEM tools, (Security Information and Event Management).

Methodology :-

The pipeline consist of 4 stages :

1. Data Collection
2. Graph Construction
3. GNN-based Detection
4. Continual learning with rehersal.

Data Collection :-

Telemetry data is collected from the Wazuh EDR Platform, including :

- Process events
- file access logs
- Network Connections
- User activites
- System events.

* Gathers real time, behavioral data from end points
* provide the raw input for attack detection.

Graph Construction :-

Telemetry data is converted into graph structures.

Ex :-

Node type	Example
Process	chrome.exe
File	config.txt
IP Address	192.168.1.5
User	admin

Edges represents interactions such as :

- Process → file access
- Process → network connection.

This forms system behavior graph.

- Converts raw logs into a structured behavior graph.
- Capture relationships between system components.

GNN - Based Detection:

After graph is built, it is processed using a GNN to detect malicious patterns.

GNN Message Passing:

- Each node updates its representation by aggregating information from neighboring nodes.
- The node representation is updated using:

$$h_v^{(k+1)} = \sigma\left(\sum_{u \in N(v)} W^k h_u^k\right)$$

h_v = node feature vector

$N(v)$ = neighbors of node

W = learnable weights

σ = activation function

During this process:

1. Nodes collect information from neighbors.
2. Multiple graph layers capture multi-stage relations
3. The network learns patterns of normal vs malicious behavior.

The GNN outputs predictions such as:

→ Benign activity

→ Suspicious activity

→ APT attack.

* Learns Behavioral Patterns of attacks
* Detects multi-stage threats.

Continual Learning with Rehearsal:

→ Cyber attacks constantly evolve, so models must adapt to new threats.

→ If we retain the model only with new data, it may suffer from catastrophic forgetting, meaning it forgets previously learned attacks.

→ To solve this, the paper uses Rehearsal-based Continual Learning.

Memory Buffer:

A small set of previous training samples is stored in mry.

This mry contains representative examples of:

→ Previous attack patterns

→ Normal behaviour.

Training Strategy:-

When new attack data arrives:

→ Train the model on new attack data.

→ Replay old samples from mry

→ Update the model using both datasets.

$$L = L_{new} + \lambda L_{replay}$$

L_{new} = loss of new data

L_{replay} = loss from stored samples

λ = balancing parameters

* Prevents catastrophic forgetting
* Adapts to evolving attack techniques
* Improves long-term detection performance.

* The main goal of this stage is to
  - maintain long-term knowledge of attacks
  - Continuously adapt to new threat patterns.

Data Collection

↓ Collect Wazuh EDR telemetry logs

Graph Construction

↓ Converts logs → behavior graphs

GNN-Based-Detection

↓ Analyze graphs to detect attack patterns

Continual Learning with rehearsal.

update model while retaining past knowledge.

Dataset :-

Benign telemetry : ~12,500 graphs

Malicious APT traces : ~8,300 graphs

Experimental Setup :

Framework : PyTorch Geometric

GPU : NVIDIA Tesla V100

3-layer GNN

Hidden Dimension : 128

Batch Size - 256

Epochs - 200

Reversal buffer - 10%

Models Compared :-

CNN BiLSTM with Attention

Static GNN

Fine-tuned GNN

Replay IDS.

Limitations :-

1. Command - and - Control (C2) detection is weaker.
   - The model performs lower on stealthy outbound communication because traffic may be encrypted or hidden.
2. Scalability issue :
   - In large enterprises, huge telemetry data increases graph size and memory usage.
3. Adversarial attacks on GNN :
   Attackers may manipulate graph structure or node features to evade detection
4. Dependence on data quality :
   Missing or incomplete logs can reduce graph accuracy.
5. Model trained on one environment may not perform equally well in another.

Results :-

Proposed model achieves F1 - Score > 0.98.

Better Recall than baseline

Minimal latency increase

Better adaptability under concept drift